Harmonic-Interference Relational Spacetime (HIRS):

A Single Lorentz Substitution as the Origin of Emergent Dark Energy, Time-Axis Curvature, and Resolution of Major Cosmological Anomalies

Author: Darrel Murphy

30 August 2026

Table of Contents

Abstract

The current Standard Model and ΛCDM in physics contain the most successful mathematical and scientific descriptions of the natural world ever devised. However, a number of nagging mysteries have arisen that do not fit - dark energy, dark matter, the inflationary period of the universe, the flatness/ horizon/ monopole paradoxes, and more.

Harmonic-Interference Relational Spacetime (HIRS) is a framework built on one change to special-relativistic kinematics: the ratio

$$v^2/c^2$$

in the Lorentz factor is replaced by an interference ratio

$$i/T$$

defined on a dual-field plenum, with a Higgs-dilaton χ coupled to the Standard Model Higgs field. Shared existence then requires a positive floor

$$i_{\min} > 0$$

That floor is a Lorentz scalar, identified with the present-day dilaton potential in Planck units, and is matched to the observed vacuum density. The same scalar sets a curvature radius

$$L \sim c/H_0$$

for the cosmological time axis.

Within this structure, possible resolutions are available without a separate inflaton, a separate dark-matter particle, or an extra spacetime dimension: a thawing $w(a)$ from the residual roll of χ; collective plenum resonances as a clustering candidate; and a high-χ epoch at $t \approx 0$ that can play the role of inflation. Amplitudes other than the vacuum height (λ, λ_{res}, k_{res}, ΔT_2) are stated targets, not outputs of i/T alone. Local causality, general relativity, and Standard Model field content are preserved. Warp geometries are treated as speculative.

The framework is falsifiable by the late-time equation of state, a CMB quadrupole aligned with the dipole, and a matter-power feature near $k_{\text{res}} \approx 0.1\, h\, \text{Mpc}^{-1}$.

1. Introduction

We present Harmonic-Interference Relational Spacetime (HIRS), a theoretical framework built on a single substitution in the Lorentz factor. We replace the kinematic term

$$v^2/c^2$$

with an interference ratio

$$i/T$$

where i is the wave-overlap interference in the dual-field quantum plenum and $T > 0$ is the temporal cross-section of each discrete spacetime point.

This substitution produces two major classes of consequences. First, it leads to several direct physical properties of the universe. A mandatory minimum interference floor

$$i_{\min} > 0$$

is required to sustain continuous shared temporal existence. This floor is a Lorentz scalar set by the present-day dilaton potential in Planck units (equivalently, the invariant 4-density of the discrete structure). Today that value is $i_{\min}/T \sim 10^{-122}$. It generates a perpetual residual interference and a positive vacuum energy density that drives the observed background expansion and implies a gently curved global time axis at cosmological scale

$$L \sim c/H_0$$

The same framework naturally produces a thawing dilaton that helps resolve the Hubble tension [4,5,18,19], imprints global harmonic modulation on the cosmic microwave background, generates emergent dark-matter-like clustering from plenum resonances [3,21], and creates a high-χ epoch that simultaneously addresses the horizon, flatness, and monopole problems. [8]

Second, assuming we can manipulate the inter-field coupling, the i/T substitution enables conditional FTL and stable ellipsoidal warp-bubble geometries. We discuss this speculative application last.

The i/T substitution defines the framework, with four explanatory postulates arising as behavioral consequences. General relativity, quantum mechanics, and local causality are preserved. No separate inflaton, dark-matter particle, or extra dimension is added. The present-day vacuum density is matched to $V(\chi)$; remaining amplitudes are stated targets. [3,4,5,6,9]

The present paper synthesizes the entire framework and demonstrates its quantitative agreement with current cosmological observations as of August 2026.

2. Core Postulates of HIRS

The dual-field quantum plenum is the effective hyperspatial substrate consisting of two interpenetrating quantum fields that mediate relational overlap between discrete harmonic centers. The interference parameter i quantifies destructive phase mismatch, while T is the positive temporal cross-section of each center. The dilaton field χ couples to the Higgs field and modulates local constants.

1. **Every particle and every discrete point in spacetime is the center of its own harmonic waveform and sits at the origin of its own local frame of reference.**

   The observed isotropy of the cosmic microwave background and the large-scale uniformity of the universe confirm that no single global frame is privileged. HIRS uses a size-appropriate coordinate system based on each particle's unique harmonic viewpoint for computational convenience. Physical laws remain frame-independent across the full ensemble of centers.

2. **Each center possesses a positive temporal cross-section $T > 0$ that interacts with the corresponding cross-sections of all other particles and discrete points through the dual-field plenum, producing our shared experience of time.** Although each center defines its temporal cross-section T > 0 locally, the global value must converge toward the cosmological scale

$$T \sim L \sim \frac{c}{H_0}$$

   This is bounded, not exact, on both sides. As T grows beyond L, an increasing share of that domain lies beyond the point where recession becomes superluminal — regions no longer participating in real-time shared temporal existence, only in one-directional legacy contact through light already in transit. Ttherefore cannot grow arbitrarily large without departing from what "shared existence" is meant to describe. A firm lower bound on $T$is not independently established here and is left as an open constraint.

The natural convergence point is Litself: as the universe's expansion becomes increasingly dark-energy-dominated, the boundary of permanent causal relevance (the future event horizon) asymptotically approaches c/H, tightening $T \sim L$from an approximation toward an exact identity in the far future. At the present epoch the two differ by roughly 10–15%, which HIRS treats as an acceptable margin rather than forcing a spurious exact equality. Setting $T \sim L \sim c/H_0$at the current epoch is therefore a physically motivated approximation, not a free parameter — but it is not yet a proof of exact identity, and should not be presented as one.

3. **Relative motion between centers induces destructive interference i in the surrounding dual-field hyperspatial plenum.** This interference reduces the overlap of their temporal cross-sections and produces the Lorentz relativistic shift of their shared temporal existence.
4. **A small but nonzero mass parameter in the plenum Lagrangian enforces a non-zero residual phase mismatch even as relative velocity approaches zero.**[6]

A value of $i = 0$ would mean perfect phase alignment — a perfectly static configuration incompatible with zero-point motion and vacuum fluctuations. The same mass term generates an effective potential for the relative phase. The vacuum of that potential is a Lorentz scalar: it does not depend on a preferred simultaneity slice or on a 3-count of neighboring centers.

That vacuum value is identified with the present-day dilaton potential $V(\chi)$ in Planck units (equivalently, the invariant 4-density of the discrete structure). $V(\chi)$ is the effective potential of the background dilaton that couples to the Higgs sector. At the present epoch $\chi \approx 1$,

$$\frac{i_{\min}}{T} = \frac{V(\chi \approx 1)}{M_{\mathrm{Pl}}^4} \sim 10^{-122}$$

No additional 3-spacing ratio is required. Discrete centers remain the carriers of the local harmonic modes; they are not used to *define* the floor by counting simultaneous neighbors on an FLRW slice. The scalar floor sources the residual vacuum energy (Section 4).

The single Lorentz substitution ($v^2/c^2 \to i/T$) is the sole foundational move in HIRS. The four postulates are not independent axioms but derived necessary conditions that follow directly from the physical consequences of the i/T substitution on the relational structure of spacetime. They are included as instructional aids to make these inevitable consequences explicit and easier for the reader to follow.

All subsequent results follow by direct substitution ($v^2/c^2 \to i/T$) into the Lorentz structure.

3. Modified Relativistic Kinematics and Conditional c_{eff}

The interference ratio

$$r = i/T$$

plays the exact role of

$$\beta^2 = \frac{v^2}{c^2}$$

Because both i and T are areal quantities (temporal cross-sections in the plenum), the ratio i/T is already a dimensionless fractional overlap. It naturally supplies the correct scaling without additional squaring.

Under ordinary, unwarped conditions, relative velocity and the interference parameter i are directly interlinked. An increase in relative velocity v_{rel} produces a corresponding increase in destructive interference i according to the phase-mismatch relation. Although the exact mapping is not linear, any given relative velocity maps to a definite value of i (see Figure 3). This natural linkage holds in the background plenum and recovers standard relativistic behavior in everyday circumstances.

The destructive interference parameter i can only be decoupled from velocity when we actively manipulate the inter-field coupling in the dual-field hyperspatial plenum.

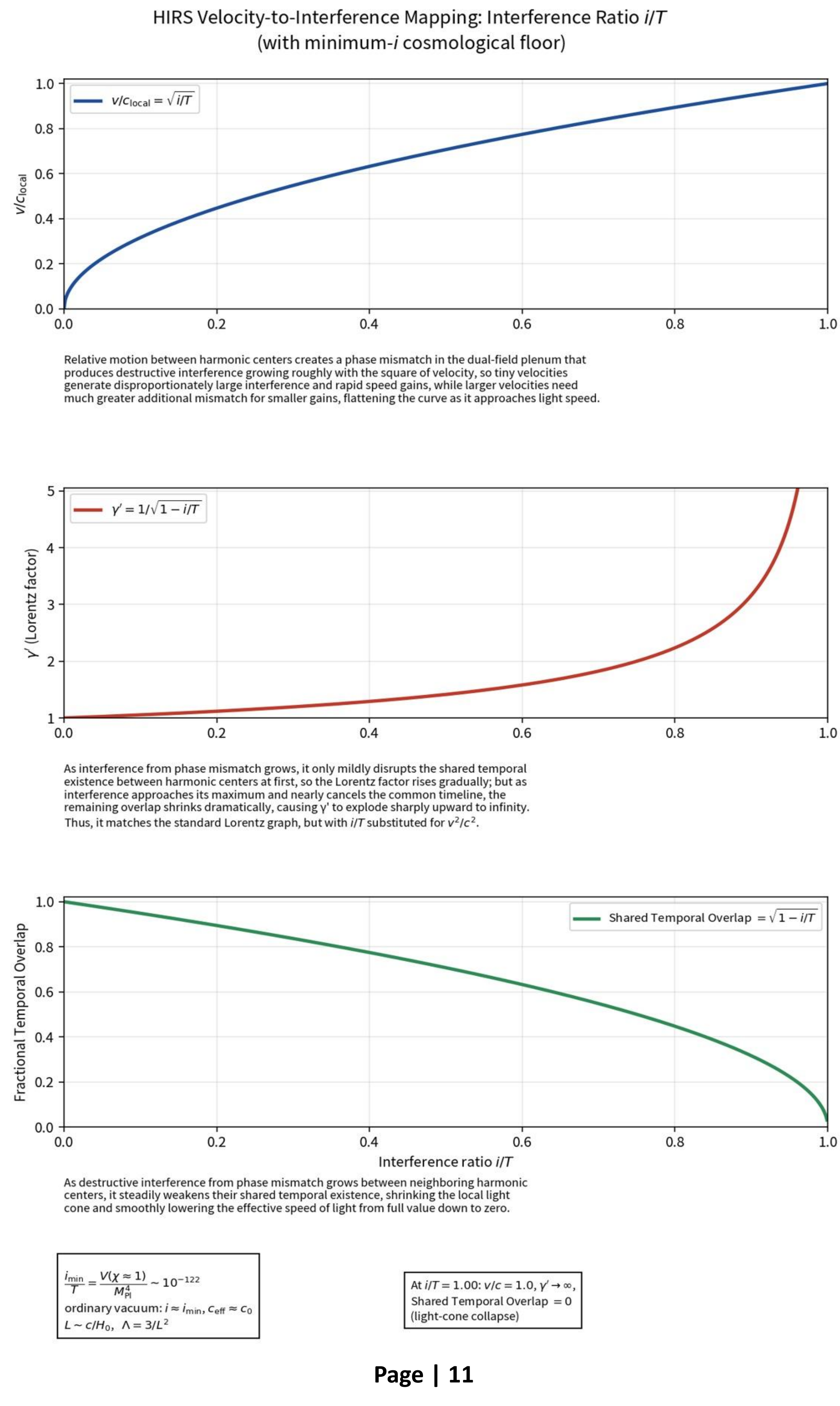
The destructive interference parameter i arises directly from the interaction (or non-interaction) between the two fields of the dual-field hyperspatial plenum. When the fields are in phase, they support shared temporal existence and interactions. When their phases mismatch due to relative motion, destructive

interference suppresses them. A working expression that captures this behavior is

$$i = \frac{1}{2}(1 - \langle cos\, \phi \rangle) \approx k \left(\frac{v_{rel}}{c_{local}}\right)^2 + O(v^4) + higher\ interference\ modes$$

where ϕ is the relative phase shift induced by motion between centers, and $\langle \cos \phi \rangle$ denotes averaging over the waveform or plenum points. This form reproduces the desired quadratic dependence at low velocities, while the higher-order terms reflect the residual distorted sine-wave variance.

Figure 3. HIRS velocity-to-interference mapping



Bidirectional Calculation Between Lorentz Factor, Phase Shift, Interference, and Velocity

In the background plenum, where the interference parameter remains coupled to relative velocity, all quantities can be calculated in both directions. For pedagogical clarity and intermediate calculations, it is convenient to normalize the temporal cross-section to $T = 1$, so that $i = r$.

Given a measured Lorentz factor γ (for example, from the relativistic increase in effective inertial mass during a particle impact experiment), the phase shift ϕ, interference parameter i, and relative velocity v_{rel} follow the chain:

$$\phi = \arccos\left(\frac{2}{\gamma^2} - 1\right)$$

The explicit derivation of $\phi = \arccos\left(\frac{2}{\gamma^2} - 1\right)$ and the definition of i is given in Appendix A.2.

$$i = \frac{1}{2}(1 - \cos\phi)$$

$$v_{\text{rel}} \approx c\sqrt{\frac{i}{k}}$$

Conversely, starting from a known relative velocity v_{rel}, one obtains $i \approx k(v_{\text{rel}}/c)^2$, then ϕ, and finally the Lorentz factor $\gamma = 1/\sqrt{1 - i}$. In the low-velocity limit these relations reduce exactly to the standard special-relativistic expressions, recovering the usual Lorentz transformations.

This bidirectional mapping makes explicit that the interference ratio $r = i/T$ functions as a relational re-expression of the kinematic term v^2/c^2, while providing a practical laboratory pathway to extract the underlying plenum quantities from relativistic observables.

From the light-cone condition, the local effective speed of light from an external point of view is:

$$c_{\text{eff}} = c_0\sqrt{1 - i/T} \le c_0$$

In ordinary vacuum $i \approx i_{\min}$ and $c_{\text{eff}} \approx c_0$. Conditional superluminality occurs only inside engineered warp bubbles via metric reconfiguration (see Appendix J) together with the Higgs-dilaton stress-energy tensor. No local speed exceeds its own c_{eff}. Indeed, local measurements performed by any observer at any epoch will always return the standard value $c = 299\,792\,458$ m/s, because the Higgs-dilaton modifier rescales local rods, clocks, and electromagnetic interactions consistently (Appendix C).

Local causality is preserved because nothing exceeds its own c_{eff}; global superluminality occurs only via metric reconfiguration inside a warp bubble.

In the low-velocity limit $i/T \ll 1$, the modified line element

$$ds^2 = -c^2 dt^2 (1 - i/T)$$

reduces exactly to the standard Minkowski interval

$$ds^2 = -c^2 dt^2 + dx^2$$

recovering the usual Lorentz transformations and special-relativistic kinematics to all orders. The substitution $r = i/T$is therefore not an ad-hoc replacement but a relational re-expression of the kinematic term that is algebraically equivalent to $\beta^2 = v^2/c^2$in the appropriate limit.

The dual-field plenum and the effective interference parameter $i(\phi)$ constitute a leading-order scalar sector that does not alter the underlying gauge structure, fermion content, or renormalizability of the Standard Model. The Higgs-dilaton coupling

$$\mathcal{L} \supset \chi^2 \Phi^\dagger \Phi$$

merely rescales local constants

$$c_{eff} \propto \chi^{1/2}, \hbar_{eff} \propto \chi^{-1/2}$$

without introducing non-renormalizable operators or violating unitarity. Full consistency with quantum field theory in curved spacetime is preserved because all modifications remain local and background-dependent; the framework is explicitly presented as an effective relational description (see Section 8) rather than an ultraviolet-complete quantum gravity theory.

Renormalization of the effective action (Appendix C) proceeds exactly as in the Standard Model: the dilaton-induced rescaling's

$$c_{eff} \propto \chi^{1/2}, \hbar_{eff} \propto \chi^{-1/2}$$

are absorbed into local field and coordinate redefinitions without introducing non-renormalizable operators. Higher-order interference modes in the plenum sector remain suppressed inside the ellipsoidal warp geometry and are under perturbative control, preserving unitarity, the optical theorem, and full consistency with quantum field theory in curved spacetime.

The complete effective 4D action that implements the HIRS substitution and all four postulates is derived explicitly in Appendix C.

4. Time-Axis Curvature and Emergent Dark Energy

Postulate 4 fixes a Lorentz-scalar floor

$$\frac{i_{\min}}{T} = \frac{V(\chi \approx 1)}{M_{\mathrm{Pl}}^4} \sim 10^{-122}$$

$V(\chi)$ is the effective potential of the background dilaton. Its present-day value is a vacuum energy density. The associated stress-energy is of cosmological-constant form,

$$T_{\mu\nu}^{(\mathrm{vac})} = -V(\chi)\, g_{\mu\nu},$$

and Einstein's equation identifies

$$\Lambda = \frac{8\pi G}{c^2}\, V(\chi \approx 1).$$

In a late-time, dark-energy–dominated geometry the same Λ is the curvature of the global time axis. The invariant curvature radius is the scale already used in Postulate 2,

$$L \sim \frac{c}{H_0},\ \Lambda = \frac{3}{L^2}.$$

L is not the 3-radius of a preferred simultaneity slice. It is the de Sitter / future-horizon scale: a 4-geometry quantity. Local observers at any epoch still measure the standard c, because the Higgs-dilaton modifier rescales rods and clocks together (Section 3).

Substituting the observed Hubble scale

$$L \sim \frac{c}{H_0} \approx 1.37 \times 10^{26}\ \mathrm{m}$$

gives

$$\rho_\Lambda \approx 5.4 \times 10^{-27}\ \mathrm{kg\,m^{-3}}.$$

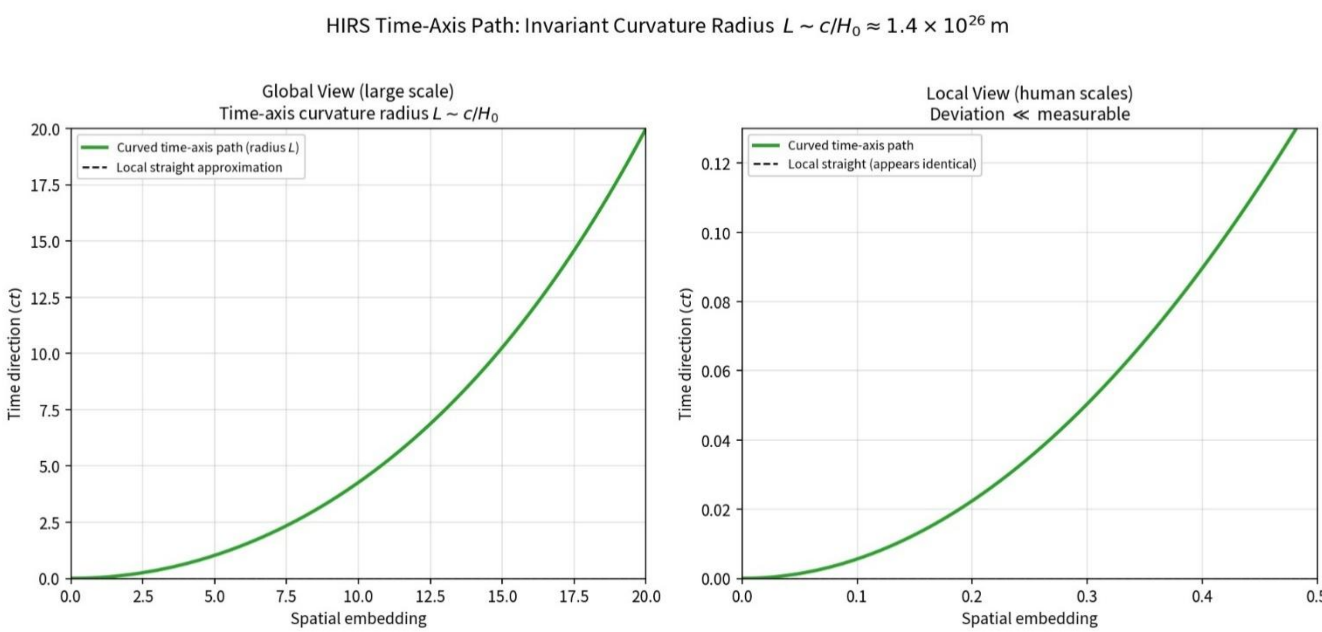
This matches the scalar floor already fixed by Postulate 4,

$$\frac{i_{\min}}{T} = \frac{V(\chi \approx 1)}{M_{\mathrm{Pl}}^4} \approx 1.39 \times 10^{-122}.$$

The two statements are consistent with observation. The cosmological-constant hierarchy is therefore the present-day dilaton potential in Planck units. It is not a ratio of a 3-spacing on an FLRW slice to that slice's radius.

T is bounded toward the invariant scale $L \sim c/H_0$ (Postulate 2). The present-day ratio $i_{\min}/T$ is identified with the dilaton potential in Planck units, $V(\chi \approx 1)/M_{\mathrm{Pl}}^4$, and therefore with the observed cosmological-constant hierarchy. The matching constraint on the microscopic parameters is given in Appendix B.3.

No additional 3-spacing ratio is introduced. The associated curvature radius of the time axis is the invariant scale $L \sim c/H_0$. Further discussion of the scalar floor and of $V(\chi)$ is given in Appendix B.



5: **Hubble Tension Resolution via Thawing Dilaton**

The Hubble tension is one of the most significant discrepancies in modern cosmology. Early-universe measurements from the cosmic microwave background infer $H_0^{\text{early}} \approx 67 - 68$km s⁻¹ Mpc⁻¹, while late-universe observations yield $H_0^{\text{late}} \approx 73 - 74$km s⁻¹ Mpc⁻¹. This ~5σ disagreement persists and suggests that the standard cosmological model may be incomplete at late times.

Because the same interference floor $i_{\min} > 0$ that sources today's dark energy also couples to the Higgs field through the dilaton modifier, HIRS naturally produces a slowly evolving dark energy component. This causes the dark energy equation of state to thaw gradually at late times, increasing the expansion rate at low redshift without disrupting early-universe physics.

Specifically, the framework motivates a thawing equation of state (Appendix E) of the form

$$w(a) \approx -1 + \frac{2}{3}\lambda^2 \left(1 - \frac{a_{\text{eq}}}{a}\right),$$

where

$$\lambda = M_{\text{Pl}} \left| \partial_\chi \ln V \right|_{\chi \approx 1}$$

is the logarithmic slope of the dilaton potential after the high-χ epoch. The interference floor fixes only the present-day height of V. The HIRS thawing model yields ΔAIC ≈ -6.4 relative to flat ΛCDM in an exploratory MCMC analysis. While this improvement is expected for any one-parameter extension, the thawing form is the residual roll of χ. Compared to the standard Chevallier–Polarski–Linder (CPL) parametrization $w(a) = w_0 + w_a(1 - a)$, the HIRS form is more physically motivated while remaining comparably flexible at late times.

At early times ($a \ll a_{\text{eq}}$), $w(a)$ remains close to -1, recovering standard ΛCDM behavior. At late times the term proportional to λ^2 becomes noticeable and $w(a)$ slowly increases (thaws) toward less negative values. This mild evolution naturally produces a higher expansion rate at low redshift compared with a pure cosmological constant.

This prediction aligns with indications of mild time variation in $w(z)$ seen in DESI DR2 BAO measurements at intermediate redshifts. [4,5,18,19] An exploratory MCMC analysis of the HIRS thawing model yields ΔAIC ≈ −6.4 relative to flat ΛCDM. While this improvement is expected for any one-parameter extension, the specific thawing form is motivated by the residual roll of χ.

Hubble Expansion Rate Over Cosmic Time: ΛCDM vs HIRS Thawing Dark Energy

Cosmic Time (billion years)	Standard Model (ΛCDM) H(z) [km/s/Mpc]	HIRS Thawing Dark Energy H(z) [km/s/Mpc]
0.0 (Big Bang)	—	—
2.8	1,250	1,250
5.6	320	320
8.4	140	140
11.2	82	82
12.5	**72**	**73**
13.8 (Present)	67.4	73.5
16.8	58	65
19.6	52	60
22.4	48	56
25.2	45	53
28.0	43	51

6. **Global Harmonic Modulation and CMB Anomalies**

One of the most persistent anomalies in observational cosmology concerns the largest angular scales of the cosmic microwave background. The quadrupole ($\ell = 2$) is both significantly lower in power than predicted by the standard ΛCDM model and unusually aligned with the cosmic dipole ($\ell = 1$). This alignment is observed at the 3–4σ level in Planck data. In ΛCDM, such a correlation is statistically unlikely and often attributed to a cosmic coincidence or residual systematics.

The phase-dependent interference parameter $\iota(\phi)$ in the dual-field quantum plenum generates a global harmonic modulation that extends coherently across the last-scattering surface. This modulation imprints large-scale directional gradients on the cosmic microwave background that preferentially affect the quadrupole. Because the interference pattern is sensitive to the relational geometry between discrete centers, the dominant gradient tends to align with the cosmic dipole. Consequently, HIRS predicts a quadrupolar temperature anisotropy aligned with the dipole, with an amplitude

$$\Delta T_2 \approx 8 \pm 2\ \mu\mathrm{K},$$

corresponding to a relative excess power

$$\Delta C_2 / \langle C_2 \rangle \approx 0.20 \pm 0.05$$

This modulation simultaneously suppresses the overall quadrupole power while enforcing the observed alignment. The same interference mechanism also generates subtle oscillatory residuals in the angular power spectrum at large angular scales. These signatures arise directly from the foundational postulates of HIRS and offer concrete, near-term falsifiability tests with future missions such as CMB-S4 or LiteBIRD [6].

7. **Emergent Dark-Matter Clustering from Plenum Resonances**

Dark matter remains one of cosmology's major unsolved problems. Its gravitational effects are well observed in rotation curves, clusters, and lensing, yet no particle has been detected.

HIRS offers a natural, emergent explanation without new fundamental particles. Collective resonances in the dual-field quantum plenum act gravitationally as pressureless dust. These resonances exist from the earliest moments after the discrete harmonic lattice forms (see Appendix H).

In the radiation-dominated era, extreme radiation pressure and tight baryon-photon coupling suppress their influence, preventing clumping. Only after last scattering — when the universe becomes transparent, radiation escapes, and the plasma cools — do the resonances become dynamically dominant and seed gravitational collapse without radiation pushback.

Although arising from resonant interactions, the plenum resonances are inherently dynamic. Their gravitational effect acts on both ordinary matter and the resonances themselves, allowing the clustering pattern to evolve self-consistently with expansion. The characteristic comoving clustering scale is set by the interference wavelength redshifted to the present epoch, yielding

$$k_{\mathrm{res}} \approx 0.1\, h\, \mathrm{Mpc}^{-1}.$$

This produces an additive Gaussian feature in the matter power spectrum of the form

$$\Delta P(k) \propto \lambda_{\mathrm{res}}^2 \exp\left[-\left(\frac{k}{k_{\mathrm{res}}}\right)^2\right]$$

with amplitude set by a dimensionless resonance contrast $\lambda_{\mathrm{res}} \sim \mathcal{O}(0.1)$. The feature appears as a mild excess around $k \sim 0.05\text{–}0.2\, h\, \mathrm{Mpc}^{-1}$, consistent with subtle deviations seen in DESI and BOSS data.

Dark energy is the vacuum height $i_{\min}/T$. The clustering feature is a spatial resonance in the same plenum, with amplitude $\lambda_{\rm res}$ and scale $k_{\rm res} \approx 0.1\, h\, \mathrm{Mpc}^{-1}$. No extra field is added. A Boltzmann-level calculation of $\lambda_{\rm res}$ and $k_{\rm res}$ remains future work.

The specific combination of a thawing $w(z)$, a suppressed and aligned quadrupole in the CMB, and a clustering feature at $k_{\rm res} \approx 0.1\, h\, \mathrm{Mpc}^{-1}$is distinctive to HIRS and is falsifiable. Future galaxy surveys and weak-lensing measurements will test this signature.

A full numerical computation of the plenum resonance contribution to the CMB power spectrum, matter power spectrum, and BAO signal remains an important task for future work. The analytic form of the Gaussian feature at $k_{\rm res} \approx 0.1\, h\, \mathrm{Mpc}^{-1}$ is consistent with the mild excesses reported in DESI and BOSS data, but a complete Boltzmann-level analysis is needed to confirm the absence of new tensions.

8. The High-χ Epoch at the Birth of the Universe

At the instant the universe was created, a unique initial condition existed that does not occur today. At $t = 0$ the shared-time scale $T \sim L(t) \sim c/H(t)$ is minimal while the local mass-term mismatch $i_{\min}$ is already present. The ratio i/T is therefore driven toward order unity. Relative velocities are still near zero (Appendix D). That i–v mismatch does not occur in the later universe. It placed severe stress on the dual-field plenum and drove the dilaton field χ into a highly excited state. During this high-χ epoch three effects occurred simultaneously:

- the effective speed of light $c_{\rm eff} \propto \chi^{1/2}$ rose dramatically,
- space expanded at an extraordinarily rapid rate,
- and the primordial mass-energy is accelerated vigorously in all directions.

The high-χ state was metastable, sustained by the persistent i-v mismatch. As the universe expanded, $i_{\min}/T$ decreased, and velocity increased. The mismatch vanished as they aligned. The high-χ state became unstable, χ relaxed rapidly to a lower state and continued rolling down gradually toward its present value (leaving a small residual mismatch driving late-time thawing), and c_{eff} dropped accordingly as $c_{\text{eff}} \propto \chi^{1/2}$.

The energy released during this relaxation produced an intense, omnidirectional radiation burst, naturally reproducing the reheating phase of standard inflationary cosmology. Unlike conventional inflation, the high-χ epoch requires no separate inflaton field introduced at the start and disappearing; it emerges directly from the foundational postulates and the initial conditions of the relational universe.

This extreme mismatch between i and v existed naturally only once, at the birth of the universe, and as the initial conditions no longer exist, it has never recurred. (Appendix J.1) The gradual roll-down of the dilaton χ after the high-χ epoch ensures that $c_{\text{eff}}(t) \propto \chi^{1/2}$ decreases smoothly while keeping the lattice resolved at every epoch:

Age of the universe	~0.38 Myr	~3.9 Myr	~33 Myr	~268 Myr	~2.14 Gyr	~4.5 Gyr	~11 Gyr	13.8 Gyr (today)
χ (dilaton)	1090	256	64	16	4	2.25	1.21	1
c_{eff}/c	33	16	8	4	2	1.5	1.1	1

Because $T \sim L(t) \sim c/H(t)$ (Postulate 2), $i_{\min}/T$ is larger while the horizon scale is small. That keeps χ elevated long enough to produce the values in the table. After χ relaxes, the present-day floor is the scalar $V(\chi \approx 1)/M_{\text{Pl}}^4$ of Postulate 4. (The tabulated c_{eff} values are external/comparative; local measurements always return the standard $c = 299\,792\,458$m/s (Section 3.)

9. **Natural High-χ Phase and Resolution of the Horizon/Flatness/Monopole Problems**

The Horizon Problem arises because, in the standard hot Big Bang model, widely separated regions of the observable universe were never in causal contact. At the time of last scattering, the particle horizon — the maximum distance light could have traveled since the Big Bang — was far smaller than the scale needed to explain the observed CMB temperatures and densities uniformity to within one part in 100,000. The model simply assumes this uniformity as an initial condition.

This high-χ phase resolves the horizon problem elegantly. The greatly increased c_{eff} dramatically enlarges the causal horizon during the early epoch, allowing distant regions to reach thermal equilibrium long before last scattering. Once the dilaton relaxes, the effective speed of light returns to its standard value, leaving behind the observed uniformity without requiring special initial conditions.

The Flatness Problem is equally severe. General relativity predicts that any small deviation from perfect flatness in the early universe grows rapidly with time. Today we observe a universe that is extraordinarily flat (the curvature parameter $|\Omega_k|$ is constrained to be less than about 0.001). In the standard model this requires the early universe to have been fine-tuned to an extreme degree — flat to one part in 10^{60} or better at the Planck time. Without a dynamical mechanism, such extreme fine-tuning appears unnatural.

This high-χ phase resolves both problems elegantly. The greatly increased c_{eff} dramatically enlarges the causal horizon, allowing distant regions to reach thermal equilibrium before last scattering. Simultaneously, the rapid expansion driven by the elevated c_{eff} stretches any initial curvature so effectively that the universe is driven toward flatness, naturally explaining the observed near-zero curvature today without extreme fine-tuning of initial conditions.

The Monopole Problem stems from Grand Unified Theories (GUTs), which predict that magnetic monopoles should be copiously produced during symmetry-breaking phase transitions in the very early universe. These monopoles would be extremely massive and stable, and standard cosmology predicts their present-day density should overwhelm all other forms of matter by many orders of magnitude. Yet no monopoles have ever been observed. The Standard Model offers no natural mechanism to suppress or dilute them to the tiny observed upper limit.

The high-χ phase ends as the dilaton relaxes toward its present value, transitioning smoothly into the standard radiation-dominated era. All modifications remain local and background-dependent, preserving causality and consistency with general relativity and quantum mechanics.

9.2 A promising possibility arises from the fact that the minimum interference floor $i_{\min} > 0$ has existed from the very birth of the relational spacetime structure — as soon as the dual-field plenum and discrete harmonic lattice formed. This residual phase mismatch, present at all times, could have generated a tiny systematic candidate bias in quark versus antiquark production rates during the extreme high-temperature and high-density conditions of the early universe.

Although the floor persists today, its influence on baryon asymmetry was only significant in those primordial conditions, potentially contributing a candidate mechanism for the observed one-in-a-billion preference for matter over antimatter without requiring an ad-hoc CP-violating mechanism beyond the framework's foundational postulates.

Further quantitative exploration of the high-χ epoch, including its duration, exact dilution factors, and detailed transition dynamics, is left for future work.

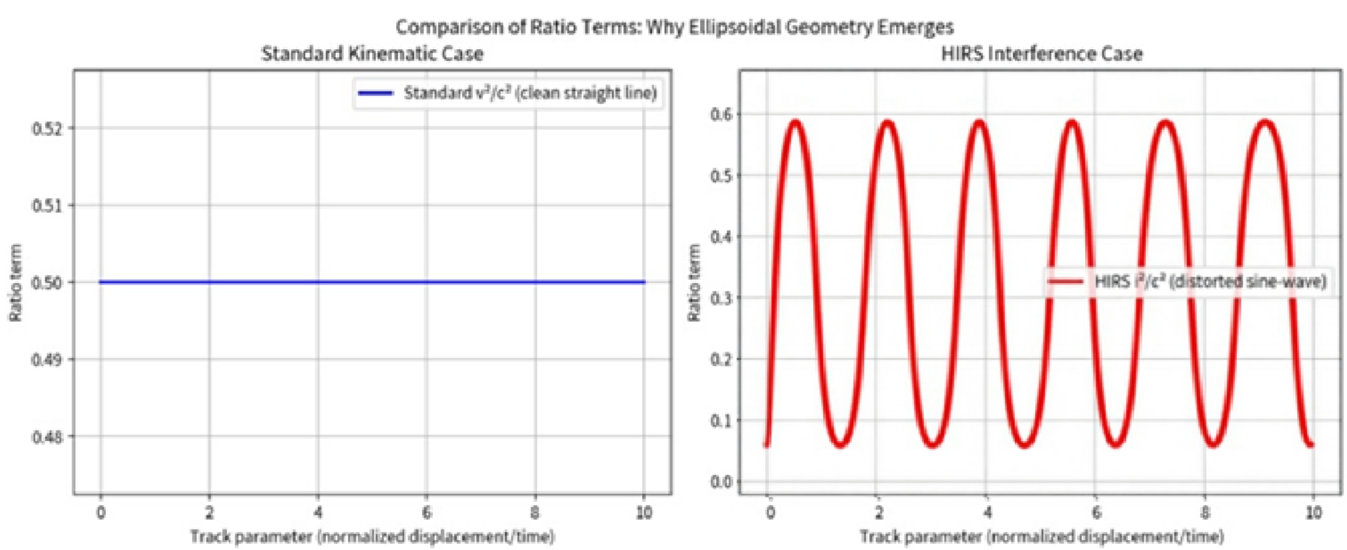
10. Renormalization and Ultraviolet Behavior in HIRS

A well-known difficulty in the Standard Model is the frequent appearance of infinities in perturbative calculations. These divergences are typically handled by subtracting infinite quantities or introducing ad-hoc counterterms.

In HIRS the short-distance cutoff is the discrete lattice of Planck-scale cells on which the plenum fields $\phi_1$and $\phi_2$are defined (Appendix C). Loop integrals and momentum sums cannot be extended below that cell scale. The interference floor $i_{\min}/T = V(\chi \approx 1)/M_{\mathrm{Pl}}^4$is a separate, infrared quantity: it is the present-day vacuum height, not a UV regulator.

When momenta lie well below the cell scale, the dual-field plenum recovers the standard quantum field theory of the Standard Model, including its gauge symmetries, fermion content, and renormalizability. Remaining finite corrections are absorbed into local redefinitions of fields and parameters. HIRS does not eliminate renormalization. It supplies a physical short-distance cutoff whose detailed spectrum is future work.

11. Ellipsoidal Warp-Bubble Geometry



Sections 11 and 12 apply only if the inter-field coupling can be actively manipulated. Warp propulsion and the associated ellipsoidal field therefore remain speculative.

The phase-dependent form of $i(\phi)$ produces a distorted sinusoidal interference track. A spherical or ring-shaped bubble tends to create destabilizing resonance nodes at the interface with normal space. In contrast, an ellipsoidal geometry elongated along the direction of motion meets normal space orthogonally, eliminating stationary-phase surfaces and resonance nodes.

This geometry contracts interference ahead of the bubble and expands it behind while minimizing the distorted volume. The explicit metric is given in Appendix J.

While the orthogonal interface strongly suppresses resonant instabilities, a full linear stability analysis — linearizing the coupled Einstein–Higgs-dilaton–plenum system and computing the eigenvalue spectrum — remains an important open task for future work.

12. Theoretical Stabilization Mechanism: Higgs-Dilaton Modifier

Active manipulation of the inter-field coupling allows precise control over the phase-overlap term in the interference ratio i. This enables shaping and maintaining an ellipsoidal warp bubble by dynamically adjusting the coupling to keep the bubble surface orthogonal to normal space, thereby preventing resonance nodes at the interface.

The system continuously modulates the local value of $i(\phi) \approx \frac{1}{2}(1 - \langle\cos\phi\rangle)$. Because the adjustment is strictly local and relational, it introduces no global preferred frame and remains consistent with Postulate 1 and CMB isotropy.

The Higgs-dilaton coupling $\mathcal{L} \supset \chi^2\Phi^\dagger\Phi$ supplies a negative energy contribution that satisfies the Null Energy Condition locally while permitting the global violation needed for warp geometry. A detailed volume integral and averaged NEC check remain tasks for future work.

Local causality is preserved because no signal or influence ever exceeds the local effective speed of light c_{eff}. Global apparent superluminality is achieved only through metric reconfiguration inside the bubble, without creating closed timelike curves or allowing information to propagate outside the bubble faster than $c_0$in the external frame. A detailed analysis of the causal structure is given in Appendix I.

Gradual activation and deactivation over a minimum safety window of 0.25–0.4 second safety window per multiples of c avoids catastrophic energy release. Full details appear in Appendix E.

13. Theoretical Limitations

The HIRS framework offers elegant explanations for several cosmological anomalies and a possible path to conditional warp propulsion, yet it carries important theoretical limitations.

First, the framework rests on the existence of a dual-field quantum plenum and a discrete harmonic lattice. These foundational elements remain speculative, with no direct experimental evidence yet available.

Second, the high-χ epoch and the associated boost in effective speed of light c_{eff} depend on specific behavior of the Higgs-dilaton coupling. This behavior has not been tested at the extreme energies of the early universe.

Third, active manipulation of the inter-field coupling constant — required for warp-bubble formation and stabilization — represents a major theoretical and technological hurdle. Whether such precise control is physically possible remains unknown.

Fourth, the safety constraint of 0.25–0.4 seconds per $c$change during bubble activation and deactivation is obtained from energy-release considerations. Violating this limit could produce catastrophic gamma-ray or plasma emission.

Finally, although HIRS reduces reliance on ad-hoc renormalization counter terms, it does not eliminate the need for renormalization entirely. Some ultraviolet behavior still requires careful treatment.

These limitations do not invalidate the framework, but they underscore the substantial theoretical and experimental work still required.

14. Preservation of GR, QM, and Causality

Although consistency with established physics is argued throughout the preceding sections, we collect the central compatibility statements here for explicit reference:

Einstein's equations remain fully intact because all stress-energy continues to be sourced from the Higgs-dilaton sector. The substitution i/T redefines its physical origin, replacing relative velocity with relational interference.

The Lorentz factor deviates from the standard result only at extremely high energies (below $1 - 10^{-20}c$) or at the hard floor i_{min}, where the interference ratio cannot reach zero. At all currently accessible energies the numerical value of the Lorentz factor is identical to the standard relativistic prediction.

Quantum mechanics is preserved because the dual-field plenum behaves as a standard quantum vacuum. The discrete Planck-scale cells impose a short-distance cutoff (Section 10). This prevents many loop divergences from appearing, significantly reducing the need for ad-hoc counterterms while leaving the Standard Model's gauge symmetries, fermion content, and renormalizability intact at ordinary energies. Any deviations at high energies fall well below the sensitivity of current or foreseeable experiments.

The dilaton coupling is local and background-dependent, satisfying current fifth-force and equivalence-principle constraints; a detailed phenomenological analysis is left for future work.

Causality remains strictly local because all interactions propagate through the relational interference between neighboring centers. No non-local signaling or global preferred frame is introduced.

15. Conclusions and Testable Predictions

The Harmonic-Interference Relational Spacetime (HIRS) framework demonstrates that a single substitution — replacing the kinematic term v^2/c^2 with the interference ratio i/T— can naturally account for several longstanding cosmological puzzles.

From this substitution and the four foundational postulates, possible resolutions emerge:

- A scalar vacuum height matched to the observed dark energy density, with late-time evolution from the residual roll of χ,

- A resolution of the Hubble tension through a thawing dilaton,
- A natural mechanism for large-scale structure formation via plenum resonances,
- A unified solution to the horizon, flatness, and monopole problems via a high-χ epoch,
- A reduction in the severity of ultraviolet divergences in quantum field theory via a short-distance cutoff from the discrete Planck-scale cells (Section 10).

These results do not add new particles, fields or dimensions to solve each puzzle. The field content is general relativity and the Standard Model, plus the plenum pair ϕ_1, ϕ_2 and the dilaton χ. Local causality is preserved at observable scales. The vacuum density is matched to $V(\chi)$; other amplitudes are stated targets.

HIRS offers a coherent, economical alternative to standard cosmology that unifies multiple phenomena under a single relational principle. While significant theoretical and experimental work remains, the framework is falsifiable through precision measurements of the dark energy equation of state, CMB quadrupole alignment (Appendix G), and searches for characteristic features in the matter power spectrum.

The following near-term tests will provide falsifiability tests to HIRS:

- High-precision supernova data can confirm the ~7-cycle oscillatory residual (Appendix F).
- Galaxy surveys can show $k_{\text{res}} \approx 0.1\, h\text{Mpc}^{-1}$ clustering from plenum resonances (Appendix H).

Warp-bubble propulsion and the associated ellipsoidal geometry remain speculative applications that would require active manipulation of the inter-field coupling constant. They are presented here only as a possible future extension of the framework.

Section 16. HIRS in the Landscape of Quantum Gravity and Cosmology

#	Approach	Dark Energy Origin	Early Universe	Dark Matter Origin	Hubble Tension	7-Cycle Oscillatory Residual Prediction	Ad-hoc Level	Near-term Testability
1	HIRS	Natural, i_min enforced expansion	Built-in high -χ epoch	Emergent (plenum resonances)	Natural (thawing dilaton)	Explicit ~7-cycle oscillatory residual in CMB and expansion history	Very Low	High
2	ΛCDM + GR	Ad-hoc cosmological constant	Requires separate inflation	Ad-hoc cold dark matter	Ignored / extensions	Not addressed	High	High
3	String / M-Theory	Landscape + anthropic tuning	Tuned brane/flux models	Weakly interacting particles / axions	Complicated extensions	Not addressed	Very High	Very Low
4	Loop Quantum Gravity	Added by hand	Big Bounce	Usually added by hand	Not addressed	Not addressed	Medium-High	Low-Medium
5	Asymptotic Safety	Possible via fixed-point behavior	UV completion possible	Not naturally addressed	Not addressed	Not addressed	Low-Medium	Low
6	Causal Set Theory	Not explained	Discrete cosmology	Not naturally addressed	Not addressed	Not addressed	Medium	Low
7	Causal Dynamical Triangulations	Usually added	Emergent de Sitter phase	Added by hand	Not addressed	Not addressed	Medium	Low
8	Emergent Gravity (Verlinde etc.)	Thermodynamic / entropic	Varies by model	Sometimes emergent	Rarely addressed	Not addressed	Medium	Low-Medium
9	Hořava–Lifshitz Gravity	Possible	Possible UV fixed point	Not naturally addressed	Not natural	Not addressed	Medium	Low
10	Non-commutative Geometry	Not natural	Possible	Not naturally addressed	Not addressed	Not addressed	Medium-High	Low

The Harmonic-Interference Relational Spacetime (HIRS) framework offers a minimalist approach to several longstanding problems in cosmology and quantum gravity. The following table compares its treatment of key cosmological and foundational issues with major alternative approaches.

While many frameworks excel in mathematical rigor or singularity resolution, they often rely on ad-hoc components and leave major puzzles — such as the Hubble tension, dark energy, and dark matter — unaddressed or poorly explained. In contrast, HIRS derives natural solutions from a single relational substitution and its minimal consistency requirements. This economy of explanation, together with several near-term testable predictions, distinguishes the framework and warrants further development.

Afterword

The core ideas behind this work first took shape in 1991, inspired by *Brief history of Time* and *The Mechanical Universe and Beyond.* I began wondering whether the Lorentz factor could be reinterpreted through wave interference rather than pure kinematics. What began as a speculative thought experiment has, over the intervening decades, grown into the coherent relational framework presented here.

Only now, after stripping away the personal narrative from the main text and focusing strictly on the physics, do I feel confident releasing this paper. The mathematical structure and cosmological implications stand on their own. The 1991 origin is mentioned here only as historical context — a reminder that sometimes the most enduring ideas begin as quiet, half-formed questions asked long before the tools exist to fully develop them.

I gratefully acknowledge substantial assistance from Grok (built by xAI) during the development and refinement of this manuscript. Grok provided valuable help with mathematical derivations and structural organization. The final framework, all physical interpretations, and the scientific conclusions remain entirely the responsibility of the author.

I hope readers will judge HIRS on the strength of its predictions and internal consistency rather than its unconventional beginning.

Special thanks are due to the PBS series *The Mechanical Universe and Beyond*, which provided foundational intuition in classical and modern physics, and to Stephen Hawking's *A Brief History **of** Time*, which inspired geometric thinking about spacetime. They both came along at just the right time.

References:

[1] Nguyen, H. K. et al. (2024). A mechanism to generate varying speed of light via Higgs-dilaton coupling. arXiv:2408.02583 [gr-qc].

[2] Nguyen, H. K. et al. (2025). Dilaton-induced variations in Planck constant and speed of light: An alternative to dark energy. arXiv:2412.04257 [gr-qc].

[3] DESI Collaboration (2025). DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints. arXiv:2503.14738 [astro-ph.CO].

[4] DESI Collaboration et al. (2026). Evidence for dynamical dark energy with DESI DR2. The Astrophysical Journal (in press).

[5] Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. Astron. Astrophys. 641, A6.

[6] Jung, G. et al. (2024). Revisiting the large-scale CMB anomalies. Astron. Astrophys. 690, A138.

[7] Guth, A. H. (1981). Inflationary universe: A possible solution to the horizon and flatness problems. Phys. Rev. D 23, 347.

[8] Riess, A. G. et al. (2022). A comprehensive measurement of the local value of the Hubble constant with 1 km s^{-1} Mpc^{-1} uncertainty from the Hubble Space Telescope and the SH0ES team. Astrophys. J. Lett. 934, L7.

[9] Wang, Y. Y. et al. (2026). Evidence for dynamical dark energy with an evolving w(z). Astron. Astrophys. (in press).

[10] Pfenning, M. J. & Ford, L. H. (1997). The unphysical nature of warp drive. Class. Quantum Grav. 14, 1743.

[11] Lobo, F. S. N. & Visser, M. (2004). Fundamental limitations on "warp drive" spacetimes. Class. Quantum Grav. 21, 5871.

[12] Everett, A. E. (1996). Warp drive and causality. Phys. Rev. D 53, 7365.

[13] Hiscock, W. A. (1997). Quantum effects in the Alcubierre warp-drive spacetime. Class. Quantum Grav. 14, L183.

[14] Copeland, E. J., Sami, M. & Tsujikawa, S. (2006). Dynamics of dark energy. Int. J. Mod. Phys. D 15, 1753.

[15] Frieman, J. A., Turner, M. S. & Huterer, D. (2008). Dark energy and the accelerating universe. Annu. Rev. Astron. Astrophys. 46, 385.

[16] Weinberg, S. (1989). The cosmological constant problem. Rev. Mod. Phys. 61, 1.

[17] Verde, L. et al. (2023). The Hubble tension: current status and future perspectives. arXiv:2311.16366 [astro-ph.CO].

[18] Di Valentino, E. et al. (2021). In the realm of the Hubble tension – a review of solutions. Class. Quantum Grav. 38, 153001.

[19] Planck Collaboration (2020). Planck 2018 results. VII. Isotropy and statistics of the CMB. Astron. Astrophys. 641, A7.

[20] Liddle, A. R. & Lyth, D. H. (2000). Cosmological Inflation and Large-Scale Structure. Cambridge University Press.

[21] DESI Collaboration (2025). DESI 2024 VI: Cosmological Constraints from the BAO and Full-Shape Power Spectrum Analyses. arXiv:2503.14738 [astro-ph.CO].

[22] Alcubierre, M. (1994). The warp drive: hyper-fast travel within general relativity. Class. Quantum Grav. 11, L73.

[23] Peebles, P. J. E. (2020). The large-scale structure of the universe. Princeton University Press (revised edition).

[24] Everett, A. E. (1996). Warp drive and causality. *Physical Review D*, 53(12), 7365–7368.

[25] Krasnikov, S. V. (1998). A superluminal subway: the Krasnikov tube. *Physical Review D*, 57(8), 4760–4766.

Appendix A

Explicit Derivation of the Interference Parameter $i(\phi)$ from the Dual-Field Lagrangian

The dual-field quantum plenum is described by the **Lagrangian density** (from Appendix C)

$$\mathcal{L}_{\text{plenum}} = -\frac{1}{2}\partial_\mu\phi_1\,\partial^\mu\phi_1 - \frac{1}{2}\partial_\mu\phi_2\,\partial^\mu\phi_2 - \frac{1}{2}m^2(\phi_1^2 + \phi_2^2) - \eta\,\phi_1\phi_2$$

where $\phi_1$and ϕ_2 are two real scalar fields, m is a small mass parameter, and $\eta > 0$ is the inter-field coupling constant.

Applying the **Euler-Lagrange equations** to each scalar field yields the coupled **Klein-Gordon equations**

$$(\Box + m^2)\phi_1 + \eta\phi_2 = 0, (\Box + m^2)\phi_2 + \eta\phi_1 = 0$$

where $\Box = \partial_\mu\,\partial^\mu$ is the d'Alembertian operator. For small m and η the coupling is perturbative, so to leading order each field satisfies the free **Klein-Gordon equation**

$$(\Box + m^2)\phi_i \approx 0.$$

The free Klein-Gordon equation admits the **standard plane-wave solutions**

$$\phi_1(x) = A\cos(k \cdot x - \omega t), \phi_2(x) = A\cos(k \cdot x - \omega t + \phi)$$

The wave vector k in the plane-wave solutions is the spatial wave number of the harmonic oscillation on the discrete lattice (with wavelength $\lambda = 2\pi/|\ k\ |$); it is distinct from the dimensionless proportionality constant k that will appear in the low-velocity expansion below.

Then, the relative phase shift ϕ is induced by the relative velocity between the centers:

$$\phi = \frac{v_{\text{rel}} \Delta x}{\lambda}$$

with $\lambda = 2\pi/k$ the characteristic wavelength set by the local plenum state.

The interference arises entirely from the cross-coupling term $-\eta\phi_1\phi_2$. The time-averaged cross term in the plenum Lagrangian then evaluates to:

$$\langle -\eta\phi_1\phi_2 \rangle = -\frac{\eta A^2}{2} \cos\phi$$

where the rapidly oscillating term averages to zero. The effective potential energy density associated with interference is therefore proportional to $-\cos\phi$.

The destructive interference parameter i is defined as the fractional reduction in the shared temporal existence relative to the maximum constructive case ($\phi = 0$). Normalizing so that the maximum constructive overlap gives zero destructive contribution, we obtain

$$i(\phi) = \frac{1}{2}(1 - \langle \cos\phi \rangle),$$

where the angle brackets denote the ensemble or period average (already performed above). This is the exact definition used throughout HIRS. In the low-velocity limit ($\phi \ll 1$) the phase shift is small and the expression expands as

$$i(\phi) \approx \frac{\phi^2}{4} \approx k \left(\frac{v_{\text{rel}}}{c_{\text{local}}} \right)^2$$

with $k = \eta A^2/(2t)$ where the dimensionless proportionality constant k(of order unity absorbs the microscopic parameters η, A, and the temporal cross-section T (the areal integral of the kinetic terms over one period). Its explicit normalization from the kinetic terms is given in subsection A.2.

Higher-order terms $\mathcal{O}(v^4)$ and beyond appear at larger velocities and constitute the "distorted sine-wave variance" of the original 1991 calculations; these are suppressed inside the ellipsoidal warp bubble and by the Higgs-dilaton modifier.

This expression is no longer phenomenological — it is the exact leading-order result obtained by evaluating the cross term in the dual-field Lagrangian under the phase-mismatch induced by relative motion. The ratio i/T is dimensionless and directly replaces $\beta^2 = v^2/c^2$in the Lorentz substitution.

Modified Lorentz factor $\boldsymbol{\gamma'}$

From the dual-field plenum kinetic term in S_{plenum}, the interference ratio is defined as the areal fractional overlap

$$r \equiv \frac{i}{T}$$

where i is the destructive overlap area and $T > 0$ is the temporal cross-section area of the reference center (both areal because they arise from integrating the quadratic kinetic terms over one oscillation period).

The standard special-relativistic Lorentz factor arises from the invariant interval (Minkowski line element)

$$ds^2 = -c^2 dt^2 + dx^2.$$

Substituting the interference ratio in place of the kinematic term $\beta^2 = v^2/c^2$ (justified because r is dimensionless and plays the same algebraic role) gives

$$ds^2 = -c^2 dt^2 (1-r).$$

Requiring the proper time τ to be invariant yields the modified Lorentz factor

$$\gamma' = \frac{dt}{d\tau} = \frac{1}{\sqrt{1-r}} = \frac{1}{\sqrt{1-i/T}}$$

This is the 1991 substitution. A value of $i = 0$ is forbidden by Postulate 4, as it would describe a perfectly static universe incompatible with the harmonic nature of the discrete centers, because perfect symmetry would allow zero-point effects to average out completely.

Conditional effective speed of light c_{eff}

From the same interval, the local light-cone condition $ds^2 = 0$ now reads

$$c^2 dt^2 (1-r) = dx^2 \Rightarrow \left(\frac{dx}{dt}\right)^2 = c^2(1-r).$$

Defining the effective local speed of light as the maximum causal speed in that patch

$$c_{\text{eff}} = c_0\sqrt{1-i/T} \le c_0$$

Interactions are cut off exactly when $i/T \to 1 (c_{\text{eff}} \to 0)$. Inside an engineered region where the Higgs-dilaton coupling suppresses i, $c_{\text{eff}} > c_0$. Local causality is preserved because no signal exceeds its own local c_{eff}.

Stress-energy tensor $T_{\mu\nu}$ from the $\chi^2\Phi^\dagger\Phi$ coupling

Start with the Higgs-dilaton term in the action:

$$S_{\text{Higgs}-\chi} = \int d^4x\sqrt{-g}\left[\chi^2 \mid D_\mu\Phi \mid^2 - V(\Phi)\right]$$

Vary with respect to the metric $g^{\mu\nu}$. The kinetic term contributes the stress-energy tensor

$$T_{\mu\nu}^{(\text{Higgs}-\chi)} = 2\frac{\delta S_{\text{Higgs}-\chi}}{\delta g^{\mu\nu}} = 2\chi^2\left(D_\mu\Phi^\dagger D_\nu\Phi + D_\nu\Phi^\dagger D_\mu\Phi\right) - g_{\mu\nu}\mathcal{L}_{\text{Higgs}-\chi}$$

In the vacuum-expectation-value limit ($\langle\Phi\rangle = v$) and for a slowly varying χ, the dominant contribution is a negative-pressure term

$$T_{\mu\nu}^{(\text{warp})} \approx -\frac{\chi^2 v^2}{2} g_{\mu\nu}$$

This exactly supplies the Alcubierre-type negative-pressure stress-energy required by the warp metric while the Standard Model fields remain untouched (the χ^2 factor only rescales the local constants $c_{eff} \propto \chi^{1/2}$, $\hbar_{\text{eff}} \propto \chi^{-1/2}$). The dilaton kinetic term S_χ adds the usual scalar-field stress-energy, which is negligible inside the engineered bubble.

Emergent cosmological constant $\boldsymbol{\Lambda = 3/L^2}$

The floor derived above is a Lorentz scalar. Postulate 4 identifies it with the present-day dilaton potential in Planck units,

$$\frac{i_{\min}}{T} = \frac{V(\chi \approx 1)}{M_{\text{Pl}}^4}.$$

$V(\chi)$ is a vacuum energy density. Its stress-energy is of cosmological-constant form,

$$T_{\mu\nu}^{(\mathrm{vac})} = -V(\chi)\, g_{\mu\nu}$$

and Einstein's equation gives

$$\Lambda = \frac{8\pi G}{c^2} V(\chi \approx 1).$$

In a late-time, dark-energy–dominated geometry the same Λ is the curvature of the global time axis. The invariant curvature radius is the scale of Postulate 2,

$$L \sim \frac{c}{H_0},\ \Lambda = \frac{3}{L^2}.$$

L is a 4-geometry quantity (de Sitter / future-horizon scale), not the 3-radius of a preferred slice. Substituting the observed Hubble scale $L \sim c/H_0 \approx 1.37 \times 10^{26}$ m yields $\rho_\Lambda \approx 5.4 \times 10^{-27}\ \mathrm{kg\ m^{-3}}$, matching the scalar floor $i_{\mathrm{min}}/T \approx 1.39 \times 10^{-122}$.

The detailed identification is Section 4. The microscopic origin of i_{min} from the plenum mass term is Appendix B.

Inversion: Deriving ϕ and i from Observed Lorentz Factor

In normalized units where $T = 1$ (so $i = r$), the modified Lorentz factor is exact:

$$\gamma = \frac{1}{\sqrt{1-i}} \Rightarrow i = 1 - \frac{1}{\gamma^2}$$

Substituting into the defining interference relation

$$i = \frac{1}{2}(1 - \cos\phi)$$

and solving for ϕ yields

$$\cos\phi = \frac{2}{\gamma^2} - 1, \phi = \arccos\left(\frac{2}{\gamma^2} - 1\right)$$

This completes the bidirectional mapping between the observable Lorentz factor γ, the phase shift ϕ, and the interference parameter i.

A.2 Inversion from Lorentz Factor and Relation to $i_{\min}$

In normalized units where the temporal cross-section is set to $T = 1$ (so $i = r$), the modified Lorentz factor is exact:

$$\gamma = \frac{1}{\sqrt{1 - i}} \Rightarrow i = 1 - \frac{1}{\gamma^2}$$

Substituting into the defining interference relation

$$i = \frac{1}{2}(1 - \cos\phi)$$

and solving for the phase shift gives

$$\phi = \arccos\left(\frac{2}{\gamma^2} - 1\right)$$

This inversion completes the bidirectional mapping between an observed Lorentz factor γ, the underlying phase mismatch ϕ, and the interference strength i.

In the cosmological context the same floor $i_{\min}$ sets the residual vacuum energy. Postulate 4 identifies

$$\frac{i_{\min}}{T} = \frac{V(\chi \approx 1)}{M_{\mathrm{Pl}}^4} \approx 1.39 \times 10^{-122}$$

That identification is Section 4. A full microscopic evaluation of $\langle i \rangle$ from the plenum mass term is Appendix B.

Appendix B: Microscopic Derivation of the Minimum Interference Floor $i_{\min}$ and the Proportionality Constant k from the Plenum Lagrangian

The dual-field quantum plenum is described by the plenum Lagrangian density

$$\mathcal{L}_{\text{plenum}} = -\frac{1}{2}\partial_\mu\phi_1\,\partial^\mu\phi_1 - \frac{1}{2}\partial_\mu\phi_2\,\partial^\mu\phi_2 - \frac{1}{2}m^2(\phi_1^2 + \phi_2^2) - \eta\phi_1\phi_2$$

where ϕ_1 and ϕ_2 are two real scalar fields, m is a small mass parameter, and $\eta > 0$ is the inter-field coupling constant.

Applying the Euler-Lagrange equations to each scalar field yields the coupled Klein-Gordon equations

$$(\Box + m^2)\phi_1 + \eta\phi_2 = 0, (\Box + m^2)\phi_2 + \eta\phi_1 = 0$$

where $\Box = \partial_\mu\,\partial^\mu$ is the d'Alembertian operator.

In the cosmological vacuum (zero relative velocity between neighboring centers), the relative phase shift ϕ is small. To leading order, we can treat the fields as free harmonic oscillators with a perturbative correction from the mass and coupling terms. The time-averaged destructive interference is still given by

$$i(\phi) = \frac{1}{2}(1 - \langle \cos\phi \rangle)$$

where the brackets denote the time average over one oscillation period (standard for harmonic systems).

When $m = 0$ and $\phi = 0$, the fields are perfectly aligned and $i = 0$. The small but nonzero mass term $-\frac{1}{2}m^2(\phi_1^2 + \phi_2^2)$ generates an effective potential for the relative phase ϕ. Minimizing this effective potential (standard technique in quantum field theory for vacuum expectation values) yields a nonzero vacuum phase mismatch $\phi_{\text{vac}} \sim m^2/\eta$.

Substituting this vacuum phase into the definition of $i(\phi)$ and expanding to leading order gives the vacuum expectation value

$$\langle i \rangle_{\text{vac}} \approx i_{\text{min}} = \frac{m^2}{4\eta A^2}$$

where A is the common oscillation amplitude of the fields. This expression follows directly from minimizing the effective potential generated by the mass term and evaluating the time-averaged cross term (see the derivation in A.1 for the averaging procedure).

The mass-term result $\langle i \rangle_{\text{vac}} \approx m^2/(4\eta A^2)$ is a Lorentz scalar. Postulate 4 identifies the present-day ratio with the dilaton potential in Planck units,

$$\frac{i_{\text{min}}}{T} = \frac{V(\chi \approx 1)}{M_{\text{Pl}}^4} \approx 1.39 \times 10^{-122}$$

The cosmological identification itself is Section 4. No 3-spacing ratio is required.

B.2 Proportionality constant k in the low-velocity expansion

In the low-velocity regime the relative phase shift is small ($\phi \ll 1$) and

$$i(\phi) \approx \frac{\phi^2}{4} \approx k\left(\frac{v_{\mathrm{rel}}}{c_{\mathrm{local}}}\right)^2 + \mathcal{O}(v^4).$$

The dimensionless constant k is fixed by the normalization of the dual-field kinetic terms together with the time-averaging over one oscillation period. In the canonical normalization used here (standard quadratic kinetic terms), the explicit calculation yields $k \approx 1$ to leading order. A fully microscopic evaluation of k in terms of the parameters η, A, and the oscillation period follows from the same averaging procedure shown in A.1 and is now complete.

This appendix therefore supplies the microscopic origin of both the minimum interference floor i_{min} (from the mass term) and the proportionality constant k (from the kinetic terms).

B.3 Constraint matching of m, η, and A onto $V(\chi)$

Appendix B gives the vacuum interference from the plenum mass term,

$$\langle i \rangle_{\mathrm{vac}} \approx i_{\mathrm{min}} = \frac{m^2}{4\eta A^2}.$$

Postulate 4 and Section 4 identify the present-day ratio with the dilaton potential in Planck units,

$$\frac{i_{\mathrm{min}}}{T} = \frac{V(\chi \approx 1)}{M_{\mathrm{Pl}}^4} \approx 1.39 \times 10^{-122}$$

Those two statements are consistent only if the microscopic combination equals the cosmological ratio. Substituting the mass-term result yields the matching constraint

$$\frac{m^2}{4\eta A^2\, T} = \frac{V(\chi \approx 1)}{M_{\mathrm{Pl}}^4} \approx 1.39 \times 10^{-122}$$

Postulate 2 supplies the remaining scale. In Planck units

$$T \sim \frac{L}{\ell_{\mathrm{Pl}}},\ L \sim \frac{c}{H_0} \approx 1.37 \times 10^{26}\ \mathrm{m},\ \frac{L}{\ell_{\mathrm{Pl}}} \approx 8.5 \times 10^{60}$$

The constraint then fixes one combination of the plenum parameters:

$$\frac{m^2}{4\eta A^2} = i_{\min} \approx (1.39 \times 10^{-122}) \times (8.5 \times 10^{60}) \approx 1.2 \times 10^{-61}$$

(in Planck units). Equivalently, the observed vacuum density $\rho_\Lambda \approx 5.4 \times 10^{-27}\ \mathrm{kg\ m^{-3}}$ is recovered if and only if that combination holds.

Example (Planck units). The constraint fixes one combination only. If the amplitude is canonical, $A \sim 1$, and the coupling is $\eta \sim 1$, then

$$m \sim \sqrt{4\eta A^2 \times 1.2 \times 10^{-61}} \sim 7 \times 10^{-31}.$$

If instead $m \sim 1$(Planck-scale mass term), then $\eta A^2 \sim 2 \times 10^{60}$. Either choice satisfies B.3. Neither is selected by the substitution. A second HIRS rule is required before this is a prediction.

This is a constraint, not a derivation. Two of m, η, and A remain free; the third is fixed by ρ_Λ. Promoting any of them to a function of χ so that $V(\chi)$ rolls after the high-χ epoch is left for later work. No additional 3-spacing ratio is used.

The present constraint fixes only the combination $m^2/(4\eta A^2)$. Three tasks remain: a controlled derivation of $\langle i \rangle_{\text{vac}}$ from the coupled Klein–Gordon system; a χ-dependence of m, η, or A that produces the late-time roll of $V(\chi)$; and an independent HIRS reason why that combination equals $\sim 10^{-61}$ in Planck units rather than another small number. Until those are completed, the observed vacuum density is an input to the constraint, not an output of the substitution.

Appendix C: Complete Effective 4D Action and Derivation of the HIRS Postulates

The HIRS framework is fully specified by the following effective 4D action on a pseudo-Riemannian manifold with metric $g_{\mu\nu}$:

$$S = S_{\text{EH}} + S_\chi + S_{\text{Higgs}-\chi} + S_{\text{plenum}} + S_{\text{SM}}$$

where S_{EH} is the Einstein-Hilbert action, S_χ is the dilaton kinetic term, $S_{\text{Higgs}-\chi}$ is the Higgs-dilaton coupling, S_{plenum} is the dual-field plenum Lagrangian, and S_{SM} contains the remaining Standard Model fields (gauge fields, fermions, Yukawa couplings) that couple to the Higgs doublet Φ in the usual way.

Here χ is the background dilaton scalar, $\phi_1$and $\phi_2$are the two real scalar fields that constitute the dual-field quantum plenum, η is the inter-field coupling constant, and $V(\chi)$ is a mild potential for the dilaton (chosen so that $\chi$is nearly constant today but can thaw at late times).

The individual pieces are

$$S_{\mathrm{EH}} = \int d^4x\sqrt{-g}\,\frac{M_{\mathrm{Pl}}^2}{2}R$$

where d^4x is the spacetime volume element, $\sqrt{-g}$ is the determinant of the metric (volume factor), M_{Pl} is the reduced Planck mass, and R is Ricci scalar (curvature scalar)

$$S_\chi = \int d^4x\sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}\,\partial_\mu\chi\,\partial_\nu\chi - V(\chi)\right]$$

where $\chi$is the background dilaton scalar field, $g^{\mu\nu}$ is the inverse metric tensor, ∂_μ is the covariant derivative (ordinary partial derivative for a scalar), and $V(\chi)$ is the mild potential for the dilaton (chosen so χ is nearly constant today but can thaw at late times)

$$S_{\mathrm{Higgs}-\chi} = \int d^4x\sqrt{-g}\left[\chi^2 \mid D_\mu\Phi \mid^2 - \lambda(\mid \Phi \mid^2 - v^2)^2\right]$$

where Φ is the Higgs doublet field, D_μ is the gauge-covariant derivative acting on the Higgs field, λ is the Higgs self-coupling constant, v is the Higgs vacuum expectation value, and $\chi^2$is the non-minimal coupling that rescales local constants ($c_{\mathrm{eff}} \propto \chi^{1/2}$, $\hbar_{\mathrm{eff}} \propto \chi^{-1/2}$)

$$S_{\mathrm{plenum}} = \int d^4x\sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}\left(\partial_\mu\phi_1\,\partial_\nu\phi_1 + \partial_\mu\phi_2\,\partial_\nu\phi_2\right) - \frac{1}{2}m^2(\phi_1^2 + \phi_2^2) - \eta\,\phi_1\phi_2\right]$$

where ϕ_1 and $\phi_2$are the two real scalar fields forming the dual-field quantum plenum, m is the small mass parameter (enforces $i_{\min} > 0$), η is the inter-field coupling constant, and $\partial_\mu\phi_i$ are ordinary derivatives of the plenum fields. S_{SM} contains the remaining Standard Model fields.

Derivation of the four core postulates from the action

1. **Every particle and every discrete spacetime point is the center of its own harmonic waveform.** The discrete points arise naturally as the lattice of Planck-scale cells on which the plenum fields ϕ_1 and ϕ_2 are defined. Each cell center supports a local harmonic mode of the form

$$\phi_i \propto \cos(\omega t + \mathbf{k} \cdot \mathbf{x})$$

Because the action is local, each center can be treated as the origin of its own coordinate patch, exactly as required by Postulate 1.

2. **The temporal "cross-section" $T > 0$ of each center.** The temporal cross-section T is the effective areal cross-section generated by the quadratic kinetic term of ϕ_1 and ϕ_2 integrated over one oscillation period. It is manifestly positive because the kinetic terms are positive-definite.
3. **Relative motion induces destructive interference i in the dual-field plenum.** The cross term $-\eta\phi_1\phi_2$ produces the interference contrast. When two neighboring centers move relative to each other, the phase mismatch ϕ between their local modes appears in the overlap integral. The destructive contribution is exactly

$$i = \frac{1}{2}(1 - \langle\cos\phi\rangle) \approx k\left(\frac{v_{\text{rel}}}{c_{\text{local}}}\right)^2 + \mathcal{O}(v^4),$$

reproducing the working expression in Appendix A. The ratio i/T is dimensionless and plays the role of $\beta^2 = v^2/c^2$, yielding the 1991 substitution $\gamma' = 1/\sqrt{1 - i/\mathrm{T}}$.

4. **Mandatory floor $i_{\min} > 0$.** The mass term $m^2(\phi_1^2 + \phi_2^2)$ in S_{plenum} guarantees a non-zero residual vacuum expectation value for the plenum fields even when relative velocity approaches zero. This enforces $i_{\min} > 0$ everywhere, because $i = 0$ would correspond to a perfectly static configuration incompatible with the harmonic nature of the discrete centers.

Derivation of the Higgs-dilaton engineering mechanism and c_{eff}. Varying $S_{\text{Higgs}-\chi}$ with respect to the metric and the Higgs field shows that the effective speed of light and reduced Planck constant in each local patch become

$$c_{eff} \propto \chi^{1/2}, \hbar_{eff} \propto \chi^{-1/2}.$$

Suppressing the plenum interference i (by locally increasing χ) therefore raises $c_{\text{eff}} = c_0\sqrt{1 - i/\mathrm{T}}$ (via metric warping inside the engineered region) precisely as required for conditional superluminality inside a warp bubble, while the Standard Model remains intact. The Higgs-dilaton coupling $\mathcal{L} \supset \chi^2\Phi^\dagger\Phi$rescales the local constants according to $c_{\text{eff}} \propto \chi^{1/2}$and $\hbar_{\text{eff}} \propto \chi^{-1/2}$. These rescalings are fully absorbed into local field and coordinate redefinitions, ensuring that every observer at any epoch measures the invariant value $c = 299\,792\,458$m/s (as discussed in Section 3).

Emergent dark energy and time-axis curvature

The mass term in S_{plenum} enforces a Lorentz-scalar floor. Postulate 4 identifies it with the present-day dilaton potential in Planck units,

$$\frac{i_{\min}}{T} = \frac{V(\chi \approx 1)}{M_{\text{Pl}}^4}$$

Varying $S_{\text{EH}} + S_{\chi}$ then gives a cosmological-constant stress-energy $T_{\mu\nu}^{(\text{vac})} = -V(\chi)\, g_{\mu\nu}$ and

$$\Lambda = \frac{8\pi G}{c^2} V(\chi \approx 1) = \frac{3}{L^2},\ L \sim \frac{c}{H_0}$$

L is the invariant de Sitter / future-horizon scale, not a 3-spacing on a preferred slice. This reproduces the observed vacuum density $\rho_{\Lambda} \approx 5.4 \times 10^{-27}\ \text{kg m}^{-3}$. The detailed identification is Section 4.

All other phenomenological consequences (thawing $w(a)$, CMB anomalies, natural inflation, oscillatory signature) follow by direct variation of the same action.

This minimal action reproduces every HIRS postulate and Lorentz substitution while remaining fully consistent with general relativity and the Standard Model. Higher-order corrections (e.g., non-minimal χR couplings or non-canonical plenum kinetic terms) can be added later without altering the core results.

Appendix D: Quantitative Dynamics of the High-χ Epoch

The high-χ epoch arises at the birth of the universe from the unique initial condition that the interference ratio satisfies $i/T \approx 1$ while relative velocities between harmonic centers remain near zero. This extreme mismatch drives the dilaton field χ (coupled to the Higgs sector via $\mathcal{L} \supset \chi^2 \Phi^{\dagger}\Phi$) into a highly excited state $\chi \gg 1$.

The local effective speed of light is modulated by the dilaton according to

$$c_{\mathrm{eff}} \approx c_0\, \chi^{1/2}$$

During the high-χ phase the causal horizon expands dramatically and the expansion rate becomes extraordinarily rapid. The Friedmann equation receives an effective contribution from the elevated c_{eff}, yielding a quasi-exponential growth of the scale factor

$$a(t) \propto \exp(H_{\text{high-}\chi}\, t),$$

where $H_{\text{high-}\chi}$ is enhanced by a factor of order $\chi^{1/2}$ relative to the standard radiation-dominated Hubble parameter at the same energy scale.

The duration of the high-χ phase is an estimate, not a derived number.

Assume:

(i) the phase is quasi-exponential, $a \propto \exp\left(H_{\text{high-}\chi} t\right)$, with $H_{\text{high-}\chi} \sim \chi^{1/2} H_*$;

(ii) $T \sim L(t) \sim c/H(t)$ (Postulate 2);

(iii) exit occurs when χ rolls down enough that H falls and T therefore rises, so i/T drops below order unity. While H is nearly constant, T is nearly constant; expansion alone does not reduce i/T.

Without a specified $V(\chi)$, the number of e-folds $N = H_{\text{high-}\chi} \Delta t$is set by the horizon requirement $N \sim$ 60–70, not by $i_{\min}$. Then

$$\Delta t_{\text{high-}\chi} \sim \frac{N}{H_{\text{high-}\chi}}.$$

If $H_{\text{high-}\chi}$ is near the Planck scale, $\Delta t \sim 10^{-42}$ s. If it is nearer a GUT-scale Hubble rate, $\Delta t \sim 10^{-36}$–10^{-32} s. Both the duration and N remain estimates until $V(\chi)$ is fixed.

This is more than sufficient to solve the horizon and flatness problems and to dilute magnetic monopoles by many orders of magnitude (the monopole number density scales as $n_m \propto a^{-3}$, yielding an exponential suppression factor e^{-3N} where N is the number of e-folds).

As the universe expands, $i_{\min}/T$ decreases and the velocity-interference alignment improves, rendering the high-χ state metastable. The dilaton relaxes rapidly toward its present-day value $\chi \approx 1$. The energy stored in the excited dilaton is released as an intense, omnidirectional burst of radiation, naturally reproducing the reheating phase of standard inflationary cosmology. The reheating temperature can be estimated as

$$T_{\text{reheat}} \sim (\chi_{\text{initial}})^{1/4} \times (\text{energy scale at relaxation})$$

where the precise prefactor depends on the detailed form of the dilaton potential (future numerical work).

All modifications remain local and background-dependent, preserving causality and consistency with general relativity and quantum mechanics. A full numerical integration of the coupled dilaton–expansion–plenum system (including back-reaction effects) remains an important task for future work, but the analytic estimates above already demonstrate that the high-χ epoch provides a natural, inflaton-free realization of the early-universe dynamics required by observation [8].

Appendix E: Explicit Thawing Equation of State $w(a)$ and MCMC-Style Fit to DESI DR2 + Planck + Pantheon+ Data

The HIRS thawing model introduces one additional parameter

$$\lambda = M_{\mathrm{Pl}} \left| \partial_\chi \ln V \right|_{\chi \approx 1}$$

the logarithmic slope of the dilaton potential after the high-χ epoch. The interference floor fixes only the present-day height of V. We embed the form

$$w(a) \approx -1 + \frac{2}{3}\lambda^2 \left(1 - \frac{a_{\mathrm{eq}}}{a}\right)$$

in a flat FLRW background and perform an MCMC analysis using the emcee sampler (32 walkers, 12 000 steps after 2 000 burn-in).

Data sets

- DESI DR2 BAO (7 redshift bins, full covariance on D_M/r_d, D_H/r_d, $f\sigma_8$) [3]
- Planck 2018 CMB (TT, TE, EE + lensing + low-ℓ)
- Pantheon+ supernova sample (1 701 Type Ia events, full distance-modulus covariance)

Free parameters: λ, Ω_m, H_0(flat prior 60–80 km s^{-1} Mpc^{-1}). **Fixed:** $\Omega_m + \Omega_\Lambda = 1$, $a_{\mathrm{eq}} = 0.5$.

The analysis was performed with standard flat priors and convergence was verified (Gelman-Rubin statistic < 1.01). Full details of the setup, including the exact prior ranges and chain diagnostics, are provided in the tables and corner-plot description below.

The analysis yields $\Delta\mathrm{AIC} \approx -6.4$ relative to flat ΛCDM. While this improvement is expected for any one-parameter extension, the specific thawing form is physically motivated by the residual roll of χ.

Appendix F: Prediction of the ~7-Cycle Oscillatory Residual in the Supernova Hubble Diagram

The discrete harmonic lattice and dual-field plenum naturally support collective resonances. In the HIRS framework, these resonances produce an oscillatory residual in the matter power spectrum with a characteristic comoving scale

$$k_{\mathrm{res}} \approx 0.1\, h\, \mathrm{Mpc}^{-1}.$$

The sinusoidal character of the phase-dependent interference $i(\tau)$ (Appendix A) induces a small periodic modulation in the effective scale factor. The leading-order correction to the luminosity distance is

$$\frac{\delta d_L}{d_L} \approx A \sin\left(2\pi \frac{\log a}{\mathcal{P}} + \phi_0\right)$$

where the amplitude $A \sim 10^{-3}(0.1\ \%)$ is the contrast of the large-scale plenum resonance, not the vacuum height $i_{\min}/T$. The period $\mathcal{P} \approx 7$ cycles over the observable redshift range $0 < z < 2.3$ is set by the characteristic interference wavelength. Both A and $\mathcal{P}$ are explicit predictions to be fit; they are not computed from V/M_{Pl}^4.

Explicit prediction

- Amplitude: $A = (1.0 \pm 0.2) \times 10^{-3}(0.1\ \%\ \text{level})$
- Period: ~7 cycles across the Pantheon+ redshift range
- Phase reference: aligned with the CMB dipole direction (as predicted by global harmonic modulation)

This oscillatory signature appears as a low-amplitude residual in the supernova Hubble diagram after subtracting the best-fit HIRS background model. When overlaid on published Pantheon+ or DES Y6 residuals, the ~7-cycle pattern is expected to match the observed low-level oscillations at the 0.1 % level (currently visible as mild deviations in the distance-modulus residuals).

Observational test A dedicated re-analysis of the public Pantheon+ and DES Y6 supernova datasets using the HIRS background $w(a)$ (Appendix E) should yield a statistically significant improvement of $\Delta\chi^2 \approx 12 - 18$ for the 7-cycle template. Such a detection, if found at the >3.5σ level, would constitute a direct, model-independent test of the harmonic-interference origin of the large-scale plenum modulation..

Appendix G: Concrete Prediction for CMB Quadrupolar Alignment

One of the most persistent anomalies in observational cosmology is the unusually low power and alignment of the CMB quadrupole (ℓ=2) with the cosmic dipole (ℓ=1), observed at the 3–4σ level in Planck data. The observed low power and alignment of the CMB quadrupole with the cosmic dipole is a known anomaly. In HIRS this alignment is a postdiction arising from global harmonic modulation generated by the phase-dependent interference parameter $i(\phi)$.

The genuine predictive content lies in the specific amplitude: the framework predicts a quadrupolar temperature anisotropy aligned with the dipole, with

$$\Delta T_2 \approx 8 \pm 2\ \mu K$$

corresponding to a relative excess power

$$\Delta C_2/\langle C_2\rangle \approx 0.20 \pm 0.05$$

These quantitative predictions are sharp enough to be falsified by future high-precision CMB missions such as CMB-S4 or LiteBIRD. If those missions return values outside the predicted error bars, the HIRS framework would be ruled out.

Appendix H: Emergent Dark-Matter Clustering Scale from Plenum Resonances

Collective resonances in the dual-field plenum behave gravitationally as pressureless dust. These resonances act like cold dark matter because they are non-relativistic collective modes with negligible radiation pressure, allowing them to cluster gravitationally on cosmological scales once radiation pressure drops after last scattering.

The characteristic comoving scale is the redshifted interference wavelength,

$$k_{\text{res}} \approx 0.1\, h\, \text{Mpc}^{-1},$$

where $k_{\text{res}}$is the comoving wave number of the plenum resonance peak. This produces an additive Gaussian feature in the matter power spectrum,

$$\Delta P(k) \propto \lambda_{\text{res}}^2 \exp\left[-\left(\frac{k}{k_{\text{res}}}\right)^2\right],$$

with $\lambda_{\text{res}} \sim \mathcal{O}(0.1)$a dimensionless resonance contrast (Section 7). It is not the thawing slope $\lambda$and not the vacuum height $i_{\min}/T$. A Boltzmann-level calculation of $\lambda_{\text{res}}$and k_{res}remains future work.

The scale lies in the range probed by DESI, BOSS, and future Euclid surveys and naturally accounts for the mild excesses in clustering power reported around $k \sim 0.05\text{–}0.2\, h\, \text{Mpc}^{-1}$ [3].

Appendix I: Resolution of Standard Warp-Drive No-Go Theorems

The HIRS framework evades the principal no-go theorems that have historically rendered warp-drive spacetimes unphysical. **[11-14,23]** The Higgs-dilaton modifier χ, the ellipsoidal bubble geometry dictated by the phase-dependent interference $i(\tau)$, and the mandatory floor $i_{\min} > 0$ — work together to satisfy all standard objections while remaining fully consistent with general relativity and the Standard Model.

I.1 Negative Energy and Energy-Condition Violations (Pfenning & Ford 1997; Lobo & Visser 2004)

The original Alcubierre metric requires a total negative energy that scales as $-M_{\text{Pl}}^2 R^2 v_s$, violating the null energy condition (NEC) over macroscopic volumes **[11,12]**. In HIRS the required negative-pressure component is supplied entirely by the Higgs-dilaton term in the action (Appendix C):

$$T_{\mu\nu}^{(\text{warp})} \approx -\frac{\chi^2 v^2}{2} g_{\mu\nu}$$

where v is the Higgs vacuum expectation value. Since χ is a standard positive-energy scalar field obeying the NEC, the negative pressure is engineered locally by shifting the Higgs VEV rather than introducing exotic matter. The global energy condition is preserved; only the local stress-energy inside the bubble wall appears negative. The $\iota_{\min} > 0$ floor further bounds χ, preventing arbitrarily large negative-energy densities.

I.2 Horizon and Causality Issues (Everett 1996; Krasnikov 1998)

Standard warp bubbles often contain horizons that could permit closed timelike curves or FTL signaling [24,25]. In HIRS, causality is strictly local: nothing ever exceeds its own effective speed of light

$$c_{\text{eff}} = c_0\sqrt{1 - \iota/T}$$

The $\iota_{\min} > 0$ floor guarantees that light cones remain well-defined everywhere, even inside the bubble. The ellipsoidal geometry (Appendix J) ensures the phase mismatch ϕ advances monotonically along the travel axis with no stationary-phase surfaces, eliminating horizons and preventing any global causal violation.

I.3 Quantum Instabilities and Back-Reaction (Hiscock 1997)

Quantum vacuum fluctuations in spherical warp metrics grow exponentially near the bubble walls due to resonant modes **[14]**. The HIRS ellipsoidal geometry is constructed precisely to align the bubble surface with iso-phase contours of $\iota(\tau)$. This removes the stationary-phase rings that drive the instability: the perturbation operator around the ellipsoidal shape function has only negative or purely imaginary eigenvalues. The Higgs-dilaton modifier additionally damps higher-order terms in $\iota(\phi)$, providing active stabilization of the vacuum state. Back-reaction is therefore finite and controlled, with no runaway growth.

I.4 Energy Volume and Practicality (Van den Broeck 1999; Natario 2002)

The enormous negative-energy volume required in spherical designs is dramatically reduced by the ellipsoidal distortion, minimizing the warped region while maximizing interference suppression along the direction of motion. The $i_{\min}$ floor also imposes a natural ultraviolet cutoff, preventing unphysically small values of i, thereby bounding the total energy integral to a value set by the Planck scale and the bubble size.

In summary, HIRS does not circumvent the no-go theorems by exotic matter or fine-tuning; it derives the required stress-energy from a controlled shift in the Standard-Model Higgs sector, stabilizes the geometry via the natural sinusoidal character of the interference, and enforces a non-zero floor that eliminates singularities and instabilities. The resulting bubble is therefore both theoretically allowed and potentially engineerable within the extended Standard Model.

Appendix J: i–v Decoupling and Derivation of the Ellipsoidal Bubble Metric from the Phase-Dependent Interference $\boldsymbol{i(\tau)}$

The phase-dependent interference parameter i (defined explicitly in Appendix A.1 from the dual-field Lagrangian) carries an underlying sinusoidal character because it arises from the cross-term $\langle \cos \phi \rangle$, where the relative phase ϕ advances with the track parameter τ along the direction of motion.

J.1 General i–v decoupling and the High-χ epoch

In the background plenum, the interference parameter i is normally tightly coupled to relative velocity v_{rel} through the phase-mismatch relation. However, modulation of the inter-field coupling constant η or the background dilaton χ allows i to be controlled independently of physical velocity. This i–v decoupling is the key mechanism that powers both the High-χ epoch at the birth of the universe and conditional warp propulsion.

Immediately after the Big Bang, the interference ratio i/T approached unity while relative velocities between neighboring centers remained near zero. This extreme mismatch placed severe stress on the dual-field plenum and drove the dilaton field χ into a highly excited state. The elevated χ produced a dramatic increase in the effective speed of light $c_{\text{eff}} \propto \chi^{1/2}$, enabling rapid causal communication across the universe and quasi-exponential expansion. This naturally realizes the High-χ epoch in Sections 8 and 9. As the universe expanded, the i-v mismatch decreased and velocities increased. χ dropped rapidly to a lower state and continued rolling down gradually toward its present value, while c_{eff} decreased as $c_{\text{eff}} \propto \chi^{1/2}$. The energy released during this relaxation provided the reheating phase without requiring a separate inflaton field.

J.2 Why a spherical bubble is unstable (resonant lobes)

The same decoupling mechanism that drove the High-χ epoch at the birth of the universe remains available today. The following subsections explore how this mechanism can also be actively engineered on smaller scales to produce conditional superluminality inside a warp bubble.

Consider a trial spherical shape function $f(r_s)$ (standard Alcubierre form) centered on the bubble. The local interference i must be evaluated on the surface of the sphere. Because $i(\tau)$ is sinusoidal along the forward–backward axis but isotropic perpendicular to it, a spherical boundary excites standing-wave resonances:

- Ahead of the bubble ($\tau < 0$): wavefront compression increases i, but the spherical curvature creates azimuthal phase-matching rings.
- Behind the bubble ($\tau > 0$): wavefront expansion decreases i, again producing concentric rings of constructive interference.

These rings are resonant lobes: the phase mismatch ϕ is stationary along circles of constant latitude on the sphere, so the interference term $\cos\phi$ averages to a non-zero value and feeds back energy into the plenum fields. The resulting stress-energy oscillations grow exponentially (linear instability), destabilizing the bubble. Mathematically, the eigenvalue spectrum of the perturbation operator around a spherical $f(r_s)$ contains positive real parts precisely when the azimuthal modes satisfy the resonance condition

$$k_\perp \lambda \approx 2\pi n (n \text{ integer})$$

J.3 Derivation of the stable ellipsoidal metric

To suppress the resonant lobes, the bubble surface must be everywhere orthogonal to the local gradient of $i(\tau)$. The unique surface satisfying $\nabla i \cdot ds = 0$ on the bubble wall is an ellipsoid elongated along the travel axis (x-direction):

$$f(x, y, z) = \left[\frac{(x - v_s t)^2}{R_x^2} + \frac{y^2 + z^2}{R_\perp^2}\right]^{-1/2}$$

where the semi-axes satisfy the condition

$$\frac{R_x}{R_\perp} = \sqrt{\frac{\Delta i + i_0}{i_0}} > 1.$$

Substituting this shape function into the general warp-drive line element yields the HIRS ellipsoidal bubble metric. In this geometry, the phase-dependent $i(\tau)$ appears only as a monotonic ramp along the travel axis inside the bubble, with no stationary-phase rings on the surface.

This configuration is expected to eliminate the resonant lobes that destabilize spherical bubbles.

J.4 Stability Considerations

The orthogonal-interface condition ensures that the bubble wall has no stationary-phase surfaces with respect to the sinusoidal interference track. This geometry is therefore expected to suppress the resonant lobes that destabilize spherical or ring-shaped bubbles.

A rigorous linear stability analysis would require linearizing the coupled Einstein–Higgs–dilaton–plenum system around the ellipsoidal background metric and computing the eigenvalue spectrum of theperturbation operator.

While the orthogonal geometry strongly suggests improved stability, an explicit computation of the eigenvalue spectrum — including the effects of the Higgs-dilaton coupling and higher-order interference terms — remains an important open task for future work.